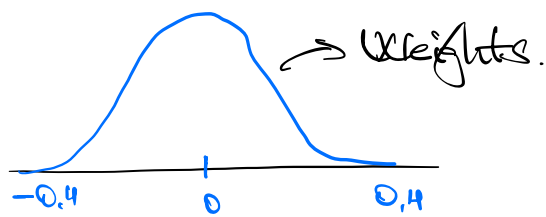
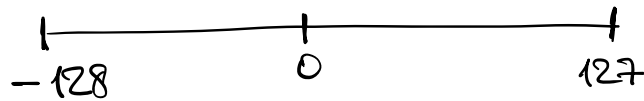


Quantization



Int 8 : 256 different birds.



We want to go from

$z_x \rightarrow z_x \rightarrow$ zero point of the input layer

$I_x \rightarrow i_x \rightarrow$ input elements of the layer

$W_x \rightarrow w_x \rightarrow$ weight element of the layer

$b_x \rightarrow b_x \rightarrow$ bias elements of the layer.

We use scales to go one to another.

For finding scales:

Weights:

We need to calculate the step size

We would like to map $[-0.4 : 0.4] \rightarrow [-128 : 127]$

$$S = \frac{\text{Float range}}{\text{Integer range}} = \frac{0.4 - (-0.4)}{127 - (-128)} = \frac{0.8}{255} = 0.00313725...$$

Note: lab document assumes weights are symmetric, so their quantization is

So

$$w_x = \text{clamp}\left(\text{round}\left(\frac{w_x}{S} + z_x\right), -128, 127\right)$$

original fp32 value

new quantized int8 value

Another way of looking at it which is how it is done in the lab manual is:

$$w_x = S_{w_x} \cdot w_x \rightarrow \text{fp32}$$

↳ the int8

$$\text{and } S_w = \frac{127}{\max(|w_x|)}$$

So if our fp32 range is $-0.48 \rightarrow 0.48$

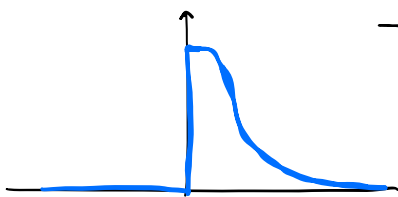
$$\text{then } S_w = \frac{127}{0.48} = 264.583$$

Meaning that to map fp32 $\rightarrow$ int8 we can multiply any fp32 by 264.583 and round it.

So if the max which is 0.48 is mapping to

$0.48 \cdot 264.583 = 127$ then smaller ones map to something else.

For Inputs (different because of RELU)



→ the weird way of doing it is not the best since we don't use anything below 0, making use of half of our offset.

Idea: Add an offset. First scale them to the full range and then center them around 0

$$i_x = S_i \cdot I_x + z_i$$

↓
new int8

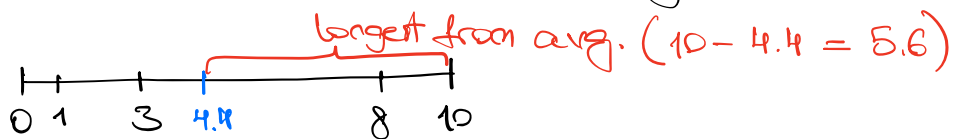
↳ old fp32

$$S_i = \frac{127}{\max(|I_x - \text{avg}(I_x)|)} \rightarrow \max(|I_x - \text{avg}(I_x)|) \text{ is the max when we plug in every element to } I_x$$

So $\max(|I_x - \text{avg}(I_x)|)$ is the farthest point from the avg value. Another way of writing is $\max(\text{avg} - 0, \max(I_x) - \text{avg})$ since the farthest point from avg is either the first one or the last.

$\max(\text{avg}, \max(I_x) - \text{avg})$ for example we have

$$[0, 1, 3, 8, 10] \rightarrow \text{avg} = 4.4$$



$$\text{So } S_i = \frac{127}{5.6} = 22.678571$$

So now, our zero point is 4.4 and to map everything from fp32 $\rightarrow$ int8 we do

$$i_x = S_i(I_x - 4.4) \rightarrow \text{so the int8 range becomes}$$

$$i_{x_{\text{first}}} = 22.678(0 - 4.4) = -99.78$$

$$i_{x_{\text{last}}} = 22.678(10 - 4.4) = 127$$

do the formulas to go from $I_x \rightarrow i_x$

$$S_i = \frac{127}{\max(|I_x - \text{avg}(I_x)|)}$$

$$i_x = S_i (I_x - \text{avg}(I_x)) \text{ expanding it}$$

$$i_x = S_i I_x - \underbrace{S_i \text{avg}(I_x)}_{z_i}$$

$$i_x = S_i I_x + z_i$$



For Biases

For biases the scale is really simple:

$$S_b = S_w \cdot S_i$$

the reason: For now disregard z_i

imagine both $i_x = S_i I_x$ & $w_x = S_w \cdot W_x$

Suppose $W = 0.5$ $S_w = 10$ so $w = 5$

$I = 2$ $S_i = 10$ so $i = 20$

And imagine $B = 1.0$

When we do $i_x \cdot w_x = 20 \cdot 5 = 100$ we get 100 instead of $2 \cdot 0.5 = 1$

Before $w \cdot I + B = 5 \cdot 2 + 1 = 11$

If we keep the bias the same after quantization, we get $w \cdot i + B = 5 \cdot 20 + 1 = 101$ which is not correct. Because the sum of bias and the

product $w \cdot i$ are not the same. To convert the bias into the unit of $w \cdot i$ we must multiply it by $S_w \cdot S_i$.

This is because bias is directly proportional to both weights and inputs. So the correct math would be

$$b_x = S_w \cdot S_i \cdot B = 10 \cdot 10 \cdot 1 = 100$$

$$Acc = 10 \cdot 5 + 100 = \underline{150}$$

Summary:

$$S_w = \frac{127}{\max(|w_x|)}$$

$$S_i = \frac{127}{\max(|I_x - \text{avg}(I_x)|)}$$

$$S_b = S_w \cdot S_i$$

$$w_x = \text{round}(S_w \cdot w_x)$$

$$i_x = \text{round}(S_i \cdot I_x + z_i)$$

$$z_i = -\text{round}(S_i \cdot \text{avg}(I_x))$$

$$b_x = \text{round}(S_b \cdot B_x)$$

Now having these equations, let's put them in our mac

$$Acc = \left(\sum_{x=0}^N I_x \cdot w_x \right) + B \quad \rightarrow \text{substitute quantized values}$$

$$w_x = w_x / S_w$$

$$I_x = (i_x - z_i) / S_i$$

$$B = b_x / S_w S_i$$

$$Acc = \left(\sum \frac{w_x \cdot (i_x - z_i)}{S_w S_i} \right) + \frac{b_7}{S_w S_i}$$

$$Acc = \frac{1}{S_w S_i} \left(\sum w_x (i_x - z_i) \right) + \frac{b_7}{S_w S_i}$$

$$Acc = \frac{1}{S_w \cdot S_i} \left[\left(\sum w_x (i_x - z_i) \right) + b_7 \right]$$

$$Acc = \frac{1}{S_w \cdot S_i} \left[\left(\sum w_x \cdot i_x - w_x z_i \right) + b_7 \right]$$

$$Acc = \frac{1}{S_w \cdot S_i} \left[\underbrace{\left(\sum w_x \cdot i_x \right)}_{\text{Combine} = b'} - \underbrace{\left(z_i \sum w_x \right)}_{b'} + b_7 \right]$$

$$Acc = \frac{1}{S_w \cdot S_i} \left(\sum w_x \cdot i_x + \underbrace{\left(b_7 - z_i \sum w_x \right)}_{b'} \right)$$

$$Acc = \frac{1}{S_w \cdot S_i} \left(\sum w_x \cdot i_x \right) + b'$$

$$\frac{Acc}{S_w \cdot S_i} = \sum w_x \cdot i_x + b'$$

→ pre computable

↳ A MAC operation

Steps (Layer 1)

- 1) Quantize weights.
find $\max(|W_{x1}|) (S_w)$
- 2) Find S_i by taking the average 1000 images
two ways to try:
 - 1) Find min-max of all 1000 individually and then average
 - 2) Find the avg of data first and then take the min-max

Get $\max(|I_x - \text{avg}(I_x)|)$ // really see if the first point or the last point is further away from avg and take the max, so $(\text{largest-value} - \text{Avg}(I_x)) \cdot S_i$ times the scale maps to 127.

Note: for actual inputs that are either >127 or <-127 when $(\text{largest-value} - \text{Avg}(I_x)) \cdot S_i$ fold them to -128 or 127 .

- 3) After each layer, we have an int32. Pass it through RELU. dequantize it by dividing the int32 by $S_w \cdot S_i$ to convert to fp32.

Requantize it again for the next layer into inte

- 4) You should precompute all the scale factors of the intermediate layers using sample data, the same way we did with the scale factors of our input. Use folding if any actual data >127 or <-128

finer steps:

Do for the first layer only for now and then see cosine similarity between the actual and what I get after dequantizing my int32

- 1) get the quantized weights in python and export.
- 2) get the quantized input for only one, and test with that

